

5 DNA replication must occur in cells before mitosis can begin.

(a) (i) Describe the role of the enzyme DNA polymerase in DNA replication.

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(ii) The two strands of a DNA molecule are described as antiparallel.

Explain the consequences of the antiparallel arrangement of the DNA strands for the replication of DNA.

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- (b) After the separation of complementary strands of DNA during DNA replication, the separated strands of DNA can become twisted and coiled together to form tangles.

The enzyme DNA topoisomerase II alpha (TOP2A) is able to remove these tangles in a two-step process:

- step 1 – TOP2A forms a complex with the tangled DNA and catalyses the cleavage (cutting) of both strands to allow the strands to separate from one another
- step 2 – TOP2A catalyses the repairs of the cuts made in the DNA so that the strands are complete.

Etoposide is a chemical that can bind to the TOP2A–DNA complex formed in step 1. This makes the complex very stable and prevents step 2 from occurring.

Suggest **and** explain why etoposide can be used to treat some types of tumour in humans.

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